

Daily Problem: March 5, 2026

Problem Statement

Construct a Turing machine that takes as input a number n in unary and accepts if and only if n is a perfect square.

Solution

Let the input be given in unary form:

$$1^n$$

where $n \geq 1$. The goal is to design a Turing machine that accepts exactly when n is a perfect square.

The construction is based on the well-known property:

$$1 + 3 + 5 + 7 + \cdots + (2k - 1) = k^2$$

Thus a number n is a perfect square if and only if it can be expressed as the sum of the first k odd numbers.

The Turing machine will repeatedly subtract consecutive odd numbers from the input. If the machine reaches exactly zero after subtracting some odd number, then the input length is a perfect square.

Step 1: Representation and Marking

The input is initially written on the tape as

$$111 \dots 1$$

with n occurrences of the symbol 1.

We use the following symbols:

- 1 : unprocessed unary symbol
- X : symbol that has been crossed out
- blank : tape blank symbol

The machine repeatedly removes blocks of size

$$1, 3, 5, 7, \dots$$

from the tape.

Step 2: Subtracting Odd Numbers

The machine maintains the current odd number k to subtract. Initially

$$k = 1$$

The machine performs the following procedure:

1. Move to the leftmost unmarked 1.
2. Cross out k occurrences of 1 by replacing them with X .
3. If fewer than k symbols remain before finishing the subtraction, reject.
4. If exactly k symbols are crossed out and no 1 remains afterward, accept.
5. Otherwise increase k by 2 and repeat.

Thus the sequence of removals is

$$1, 3, 5, 7, \dots$$

which corresponds to subtracting successive odd integers.

Step 3: Correctness

Suppose the input length is n .

After subtracting the first k odd numbers we have removed

$$1 + 3 + 5 + \cdots + (2k - 1)$$

symbols. By the identity

$$1 + 3 + 5 + \cdots + (2k - 1) = k^2$$

the total removed equals k^2 .

If the machine reaches exactly zero remaining symbols after subtracting an odd number, then

$$n = k^2$$

and the machine accepts.

If at some stage the machine attempts to remove more symbols than remain, then

$$n \neq k^2$$

for all integers k , and the machine rejects.

Conclusion

The Turing machine works by repeatedly subtracting consecutive odd numbers from the unary input. Since the sum of the first k odd numbers equals k^2 , the machine reaches zero exactly when the input length is a perfect square. Therefore the machine accepts if and only if the input 1^n represents a perfect square.